3-Day Literature Collection Plan

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Includes all Claude Code prompts, API setup, and execution scripts

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Executive Summary

This plan supersedes the ChatGPT/Gemini framework in one critical dimension: it adds **Papers With Code API** as a primary source (giving pre-curated, benchmark-evaluated papers instantly), uses **Semantic Scholar batch endpoints** to cut API time from days to hours, and mines **arXiv Related Works sections** as expert-curated seed expansion. The goal is 150–200 high-quality, matrix-ready papers by end of Day 3 — enough for a deep, journal-accepted survey.

> **★ Three additions missing from the ChatGPT/Gemini plan**
> 1) Papers With Code API — structured task pages with leaderboards, pre-tagged by dataset and metric. 2) Semantic Scholar /paper/batch endpoint — 500 papers per call instead of 1, cutting runtime by ~98%. 3) arXiv Related Works mining — each seed paper's reference list is expert-curated signal, far more targeted than raw citation graphs.

Why this plan is stronger

Dimension	ChatGPT / Gemini plan	This plan
Papers With Code	Not mentioned	Primary source — pre-tagged, leaderboard-linked
Semantic Scholar API	1 paper per call (slow)	/paper/batch — 500 per call, free key = 1,000 req/5min
arXiv mining	Pull latest preprints only	Mine Related Works sections as expert-curated expansion
Target paper count	1,000+ (quantity focus)	150–200 high-quality (journal cares about coverage, not count)
Seed API key	Not mentioned	Get free Semantic Scholar key first — 10x rate limit
Filtering method	Script only	Script + Claude chat batch filtering (paste 20–30 abstracts at a time)

1 Day 1 — Seed Collection + Environment Setup

Manual Google Scholar (2–3 hrs) · API keys · seed_papers.csv ready by evening

Step 1.1 — Get your free API keys (do this first, takes 5 minutes)

Semantic Scholar unauthenticated = 100 requests per 5 minutes. With a free key = 1,000 requests per 5 minutes. This is a 10x speedup that alone makes the difference between a 2-day script run and a 4-hour one.

- Semantic Scholar key: go to **semanticscholar.org/product/api** → Request API Key (free, instant)
- OpenAlex: no key needed — just add your email as **mailto:youremail@domain.com** in the API query string for a polite-pool rate boost
- Papers With Code: no key needed — fully open REST API
- arXiv: no key needed — open API

Step 1.2 — Manual Google Scholar seed collection (2–3 hours)

Run the following targeted queries. For each query: inspect top 20 relevance-sorted results + top 10 date-sorted results only. Do not read papers — just save title, DOI, venue, year, and modality into your spreadsheet. Target 60–70 seeds total.

■ Surveys & taxonomy (target: 5–10 papers)
"class-agnostic counting" survey | "visual counting" survey | "object counting" "computer vision" survey | ...

- ■ Search: "class-agnostic counting" survey
- ■ Search: "visual counting" survey
- ■ Search: "object counting" "computer vision" survey
- ■ Search: "few-shot object counting" taxonomy
- ■ Search: "open-vocabulary object counting" survey

■ Image counting (target: 20–30 papers)
"few-shot object counting" | "exemplar-based object counting" | "class-agnostic object counting" | ...

- ■ Search: "few-shot object counting"
- ■ Search: "exemplar-based object counting"
- ■ Search: "class-agnostic object counting"
- ■ Search: "zero-shot object counting"
- ■ Search: "open-vocabulary object counting"
- ■ Search: "object counting" "FSC-147"
- ■ Search: "object counting" "FSCD-147"

- Search: "object counting" SAM
- Search: "object counting" "foundation models"
- Search: "visual counting" "vision-language"

Video counting (target: 10–15 papers)
"video object counting" | "open-world object counting in videos" | "unique instance counting" video | ...

- Search: "video object counting"
- Search: "open-world object counting in videos"
- Search: "unique instance counting" video
- Search: "TAO-Count" object counting
- Search: "VideoCount" object counting
- Search: "CountVid" object counting
- Search: "video counting" "re-identification"

Extended modalities (target: 10–15 papers)
"3D object counting" | "object counting" "point cloud" | "RGB-D object counting" | ...

- Search: "3D object counting"
- Search: "object counting" "point cloud"
- Search: "RGB-D object counting"
- Search: "remote sensing" "object counting"
- Search: "cell counting" "computer vision"
- Search: "aerial image" "object counting"
- Search: "thermal image" "object counting"
- Search: "event camera" "object counting"

Datasets & benchmarks (target: 10–15 papers)
"FSC-147" object counting | "FSCD-147" object counting | "CountBench" object counting | ...

- Search: "FSC-147" object counting
- Search: "FSCD-147" object counting
- Search: "CountBench" object counting
- Search: "OmniCount" object counting
- Search: "ShanghaiTech" counting
- Search: "CARPK" counting
- Search: "Science-Count" object counting

Step 1.3 — Save seeds to seed_papers.csv

Your CSV must have exactly these columns. Claude Code will read this file on Day 2:

```
title, authors, year, DOI, venue, modality, URL, notes

# modality options: image | video | 3D | remote-sensing | microscopy | other
# Leave DOI blank if unknown — script will resolve it via Crossref
# URL: paste the Google Scholar or arXiv link
# notes: anything you noticed (e.g. 'highly cited', 'key benchmark paper')
```

■ Day 1 stopping rule

Stop manual Google Scholar when you hit 60 seeds OR after 3 hours — whichever comes first. Do not spend more time on manual collection. The script handles everything else. Quality of seeds matters more than quantity — 40 strong seeds beats 80 weak ones.

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Day 2 — Automated Expansion (Run Overnight → Filter by Mor

Papers With Code · Semantic Scholar batch · OpenAlex · arXiv Related Works

Step 2.1 — Get your API key (if not done on Day 1)

```
# Set your Semantic Scholar API key as environment variable
export S2_API_KEY='your_key_here'

# Install required Python packages
pip install pandas requests openpyxl fuzzywuzzy python-Levenshtein tqdm
```

Step 2.2 — The complete Claude Code prompt (paste this verbatim)

Open Claude Code in your terminal and paste the following prompt exactly. Claude Code will write and execute the full pipeline script for you:

> ■ **Claude Code Prompt — Copy and paste this exactly**
> The prompt below instructs Claude Code to build the complete pipeline. Copy everything between the dashed lines.

Step 2.2 (continued) — Full Claude Code prompt, part 1 of 2

```
I am writing a survey paper on object counting across modalities.
I have a file called seed_papers.csv with columns:
  title, authors, year, DOI, venue, modality, URL, notes

Build a complete Python pipeline that does the following:

=== SOURCE 1: Papers With Code API (run this first) ===
Query: https://paperswithcode.com/api/v1/tasks/?q=object+counting
- Retrieve all papers linked to object counting tasks
- Also query: https://paperswithcode.com/api/v1/datasets/
  for FSC-147, FSCD-147, CountBench, OmniCount, VideoCount,
  TAO-Count, ShanghaiTech, CARPK, OmniCount
- Extract: paper title, authors, year, arxiv_id, URL, dataset, metrics reported
- Save to pwc_papers.csv

=== SOURCE 2: Semantic Scholar batch endpoint ===
Use the /graph/v1/paper/batch endpoint (NOT single-paper endpoint):
  POST https://api.semanticscholar.org/graph/v1/paper/batch
  Headers: {x-api-key: YOUR_S2_KEY}
  Body: {ids: [list of DOIs or S2 paper IDs], fields:
    'title,authors,year,venue,abstract,citationCount,
     references,citations,influentialCitationCount'}
- Process seeds in batches of 500
- For each seed: get all citing papers (forward lookup)
- For each seed: get all referenced papers (backward lookup)
- Cap expansion at 2,000 candidate papers total
```

Step 2.2 (continued) — Full Claude Code prompt, part 2 of 2

```
=== SOURCE 3: OpenAlex API ===
For each seed DOI query:
  https://api.openalex.org/works?filter=doi:{DOI}
  &select=title,authorships,publication_year,primary_location,
  abstract_inverted_index,cited_by_count,referenced_works
  &mailto=youremail@domain.com
- Extract cited_by and references for each seed
- Merge with Semantic Scholar results, dedup by DOI

=== SOURCE 4: arXiv Related Works mining ===
For each seed that has an arXiv ID:
  Query: https://export.arxiv.org/abs/{arxiv_id}
  Parse the HTML to extract the References section
  Extract all paper titles mentioned in the references
  Then search each title on Semantic Scholar to resolve metadata
- This gives expert-curated signal beyond citation graphs

=== DEDUPLICATION ===
- Primary key: DOI (exact match)
- Secondary: fuzzy title match with Levenshtein similarity > 0.85
- Keep the record with more metadata when deduplicating

=== FILTERING ===
Apply these inclusion rules automatically:
INCLUDE if abstract contains ANY of:
  ['counting', 'density map', 'MAE', 'RMSE', 'exemplar',
   'few-shot count', 'class-agnostic', 'object count',
   'crowd count', 'cell count', 'unique instance']
EXCLUDE if abstract contains ONLY generic detection terms
  and no counting-specific evaluation (no MAE/RMSE/OBO/Anchor-AP)

=== OUTPUTS ===
1. literature_matrix.xlsx with columns:
   title | authors | year | venue | DOI | URL | modality |
   task_type | input_type | output_type | datasets | metrics |
   key_contribution | citation_count | relevance_score |
   bibtex_copied | pdf_saved | survey_section | notes

2. references.bib — BibTeX for all included papers
   Escape all special characters for LaTeX compatibility
```

```
   Use DOI-based keys: author_year_firstword format

3. missing_metadata_report.csv — papers with missing DOI/abstract

4. grouped_papers.md — papers grouped into these themes:
   - survey/taxonomy
   - density-map/regression-based
   - detection-based counting
   - few-shot/exemplar-based
   - class-agnostic counting
   - zero-shot counting
   - open-vocabulary/foundation-model
   - video object counting
   - 3D/point-cloud/depth
   - remote-sensing/aerial/UAV
   - microscopy/cell/scientific
   - datasets and benchmarks
   - applications

5. survey_outline.md — draft survey outline with sections,
   subsections, and 3-5 suggested papers per subsection

Add a progress bar (tqdm) and print a summary at the end:
  Total candidates found | After dedup | After filtering | By modality
```

Step 2.3 — Run the script and let it work overnight

Start the script before you go to sleep. With the Semantic Scholar batch endpoint and your free API key, expect a runtime of 3–5 hours for full expansion from 60 seeds. Do not interrupt it.

> ■ **Expected runtime with API key**
> Papers With Code fetch: ~5 minutes. Semantic Scholar batch (500/call): ~30–60 min for full expansion. OpenAlex: ~60 min. arXiv mining: ~30 min. Total: 3–5 hours. Without the API key (100 req/5min), the same job takes 40+ hours — so the key is non-negotiable.

Step 2.4 — Morning filtering (while coffee is brewing)

By morning you will have 1,200–2,000 candidate records in literature_matrix.xlsx. Your filtering workflow:

- Open literature_matrix.xlsx and sort by **relevance_score** descending
- Papers scored high (script auto-filtered) → keep, spot-check 10% of them
- Papers scored medium → paste abstracts into Claude chat in batches of 20–30 and ask:

```
# Paste this into Claude.ai chat with your abstracts:

Here are 25 paper abstracts. For each one, tell me:
KEEP or REMOVE
Reason: one sentence

Inclusion criteria:
- Directly proposes a method, dataset, or metric for object counting
- Evaluates against MAE, RMSE, OBO, Anchor-AP, or similar counting metrics
- Covers image, video, 3D, remote sensing, microscopy, or multi-modal counting

Exclusion criteria:
- Generic object detection (YOLO/Faster R-CNN) with no counting evaluation
- Counting mentioned only as a downstream application, not the core contribution

[PASTE YOUR ABSTRACTS HERE, numbered 1–25]
```

- Target: reduce to **150–200 high-quality papers** in the matrix by end of Day 2

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Day 3 — Gap-Filling, BibTeX Cleanup, Matrix Finalisation

Targeted arXiv queries · missing metadata · LaTeX validation

Step 3.1 — Check modality coverage

Citation graphs are biased toward heavily-cited image counting papers. Video, 3D, and event-camera papers are underrepresented. Check your matrix counts:

Modality	Minimum target	If below target: run this arXiv query
Image (few-shot / open-vocab)	50–70 papers	search arXiv cs.CV for 'few-shot object counting' + 'class-agnostic counting'
Video counting	15–20 papers	search arXiv cs.CV for 'video object counting' + 'unique instance counting'
3D / point cloud	8–12 papers	search arXiv cs.CV + cs.RO for '3D object counting' + 'point cloud counting'
Remote sensing	8–12 papers	search arXiv cs.CV for 'aerial object counting' + 'UAV counting'
Microscopy / cell	6–10 papers	search arXiv cs.CV + q-bio for 'cell counting deep learning'
Surveys / benchmarks	10–15 papers	manually check 'Cited by' chains from your top survey seed papers

Step 3.2 — arXiv gap-fill prompt for Claude Code

Paste this into Claude Code for any modality that is below target:

```
# Claude Code prompt for arXiv gap-filling

Search arXiv using the API for papers on [MODALITY] object counting.
Query URL format:
  http://export.arxiv.org/api/query?search_query=
    ti:[search terms]+AND+cat:cs.CV
    &sortBy=relevance&max_results=50

Run these queries (replace [MODALITY] with your target):
  'video object counting'
  'unique instance counting video'
  '3D object counting point cloud'
  'aerial object counting UAV'

For each result:
- Extract: title, authors, year, arXiv ID, abstract, DOI if available
- Check against existing literature_matrix.xlsx by fuzzy title match
- Add only papers NOT already in the matrix
- Append to literature_matrix.xlsx with modality column filled
- Update references.bib with new BibTeX entries
- Print: X new papers added for [MODALITY]
```

Step 3.3 — Resolve missing DOIs

```
# Claude Code prompt for DOI resolution

Read missing_metadata_report.csv.
For each paper with a missing DOI:
1. Query Crossref API:
   https://api.crossref.org/works?query.title={TITLE}
   &query.author={FIRST_AUTHOR}&rows=3
2. Take the top result if title similarity > 0.85
3. Extract DOI and update literature_matrix.xlsx
4. Generate a BibTeX entry and append to references.bib
5. Print a summary: X DOIs resolved, Y still missing

For papers still missing DOI after Crossref:
- Try Semantic Scholar search by title
- If still not found, flag as 'DOI_UNRESOLVED' in matrix
```

Step 3.4 — Validate BibTeX for LaTeX

```
# Claude Code prompt for BibTeX validation

Read references.bib and validate every entry:

1. Check for unescaped special characters in all fields:
   & -> \&    % -> \%    $ -> \$    # -> \#    _ -> \_
   { and } must be balanced

2. Check all required fields are present:
   @article: author, title, journal, year, volume OR pages
   @inproceedings: author, title, booktitle, year
   @misc (arXiv): author, title, year, eprint, archivePrefix

3. Check for duplicate BibTeX keys — make each unique
   Key format: FirstAuthorLastname_Year_FirstTitleWord

4. Fix all issues and save as references_validated.bib
5. Print validation report: X entries checked, Y fixed, Z warnings
```

Step 3.5 — Generate the semantic clustering + survey outline

Run this final Claude Code prompt to generate your thematic grouping and draft outline:

```
# Claude Code prompt — final taxonomy + outline generation

Read literature_matrix.xlsx (final filtered version).

=== CLUSTERING ===
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.cluster import KMeans

Use abstract + title as text. TF-IDF, max_features=3000.
K-Means with n_clusters=8, random_state=42.
Label each cluster by its top-10 TF-IDF terms.
Map cluster IDs to human-readable theme names.
Add cluster_label column to matrix.

=== OUTPUTS ===
1. grouped_papers_final.md
   For each theme: list papers with year, venue, one-line contribution

2. survey_outline_final.md
   Generate a full survey paper outline with:
   - Section number and title
   - 2-sentence description of what the section covers
   - 3-5 key papers to cite in each section (by BibTeX key)
   - Suggested table or figure for each major section

3. benchmark_compendium.md
   Table of all datasets found in matrix:
   Dataset | Modality | Size | Input prompt | Output | Key metrics | SOTA method

4. collection_summary.md
   Total papers | By modality | By task type | By year (histogram) |
   Top 10 most-cited | Papers missing PDF | Recommended reading order
```

✓ Day 3 done — what you should have by end of day

literature_matrix.xlsx with 150–200 papers, all fields filled. references_validated.bib compiling cleanly in LaTeX. grouped_papers_final.md as your writing roadmap. survey_outline_final.md as your paper skeleton. benchmark_compendium.md as your dataset section draft. You are now ready to begin writing.

Quick Reference — All API Endpoints

API	Key required?	Key endpoint	Best for
Papers With Code	No	paperswithcode.com/api/v1/tasks/?q=object+counting	Pre-tagged benchmark papers
Semantic Scholar	Free key = 10x rate	api.semanticscholar.org/graph/v1/paper/batch	Citation graphs, abstracts
OpenAlex	No (use mailto=)	api.openalex.org/works?filter=doi:{DOI}	Bulk metadata, open access
arXiv	No	export.arxiv.org/api/query?search_query=...	Latest preprints, Related Works
Crossref	No	api.crossref.org/works?query.title={TITLE}	DOI resolution

Quick Reference — Literature Matrix Fields

Field	Valid values / format	Purpose
modality	image | video | 3D | remote-sensing | microscopy | thermal | event-camera | other	Data input type
task_type	density | detection | few-shot | zero-shot | open-vocab | tracking	Algorithm mechanism
input_type	bboxes | point-exemplars | text | class-names | unsupervised	Inference-time conditioning
output_type	scalar | density-map | bboxes | masks | trajectories	Localization level
metrics	MAE | RMSE | OBO | Grid-MAE | Anchor-AP | MOTA | ID-Switches	Benchmark standard
relevance_score	high | medium | low	Filter priority
survey_section	Section number from outline (e.g. 3.2)	Writing organisation

3-Day Timeline Summary

Day	Morning	Afternoon / Evening	Overnight	Target output
Day 1	Get API keys (5 min). Start Google Scholar seed search.	Complete seed search. Build seed_papers.csv.	Set up Claude Code. Paste Day 2 prompt. Start script.	60–70 seeds in CSV. Script running.

Day	Morning	Afternoon / Evening	Overnight	Target output
Day 2	Script finishes. Check summary output.	Batch-filter abstracts in Claude chat. Reduce to 150–200 papers.	Run clustering + outline generation prompt.	literature_matrix.xlsx filtered. references.bib draft.
Day 3	Check modality coverage. Run arXiv gap-fill if needed.	Resolve missing DOIs. Validate BibTeX.	Review grouped_papers_final.md and survey_outline_final.md.	Complete matrix. Clean BibTeX. Survey outline ready.

→ When you are ready to write

Come back to Claude and share your grouped_papers_final.md and survey_outline_final.md. We will then work on the scholarly argument — the historical arc, open challenges framed as paradoxes, the benchmark reproducibility audit, and the way-forward section that journals reward. The collection phase you are doing now is the foundation. The writing is where it becomes award-winning.